CIS241

System-Level Programming and Utilities

SSH

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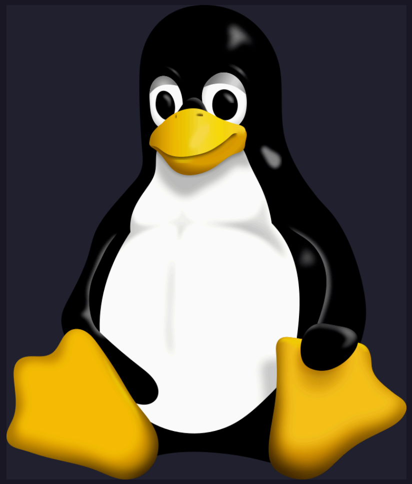
> **Based on material provided by Erin Carrier, Austin Ferguson, and Katherine Bowers**

First, Linux



What is it?

- **Often referred to as an operating system (OS)**
  - Competing with Mac/Windows
- ***Technically*, it is a kernel**

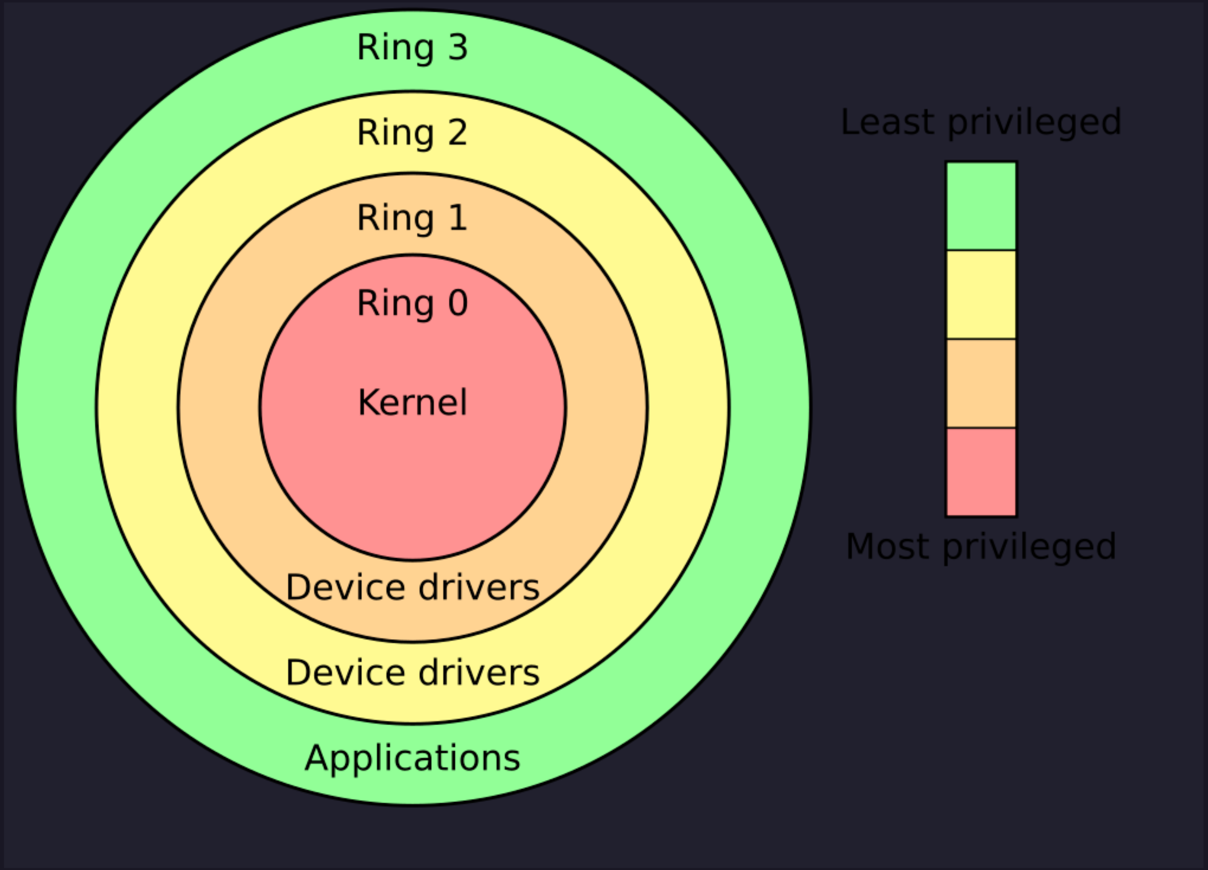
Many OSs use the Linux kernel

- Ubuntu (what is on EOS)
- Arch, Kali, Gentoo, Fedora, etc.

Isn't Mac Linux too?

- Nope, Darwin! (Unix-based)

Protection Rings



A bit of history

1970s: UNIX source code typically stripped from distribution

- Exorbitant amounts of money to acquire source

Linux first developed by Linus Torvalds to be a copy of UNIX

- **Released it to Internet for suggestions on how to improve it**
- **Kernel only updated by core developers and from suggestions**
- **Short history of Linux:**
  - https://cvw.cac.cornell.edu/linux/intro/history

Unix?

Unix was an earlier operating system

- Linux and other systems are heavily influenced by Unix

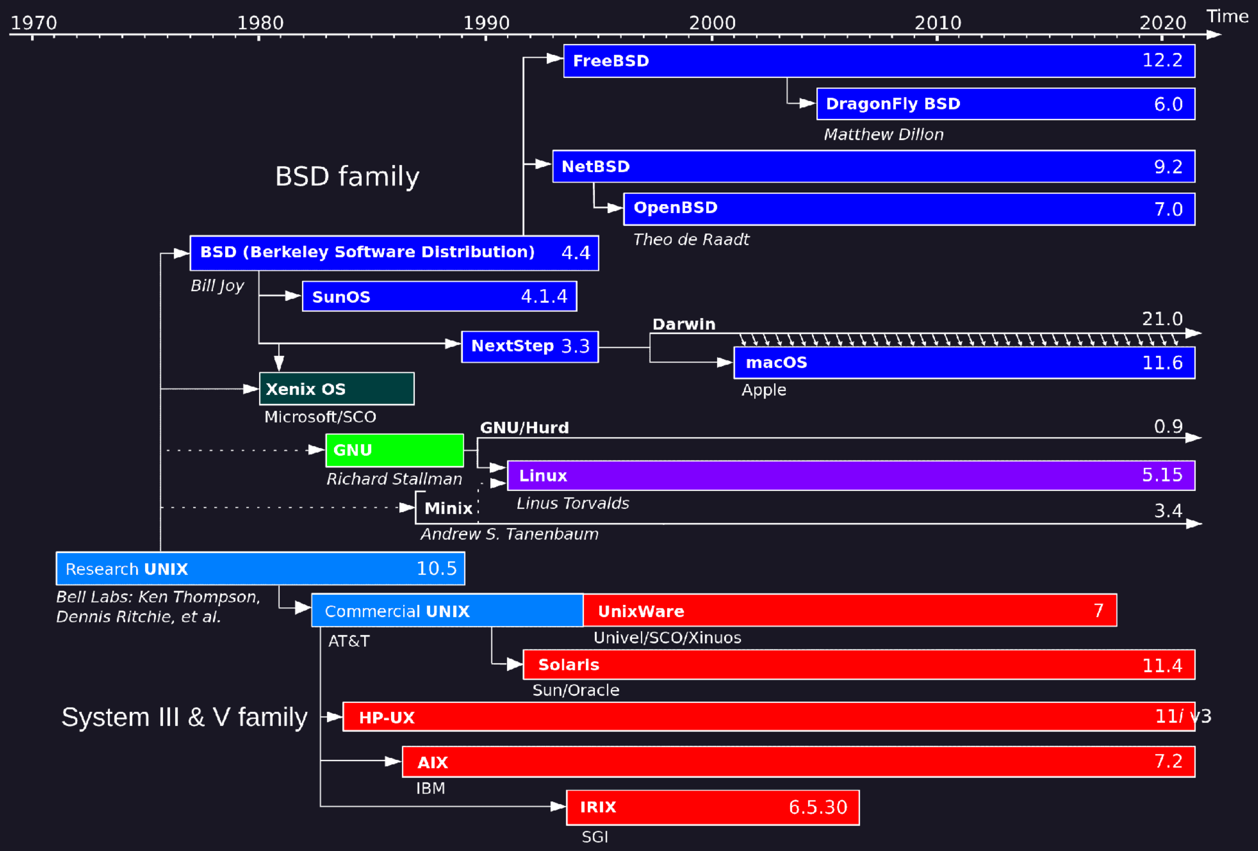
We now use “Unix-like” to describe systems

- They follow the same general ideas as Unix
- This means most things (commands, properties) carry over!

"Unix-like" systems:

- Linux
- macOS
- FreeBSD
- Solaris

Unix Timeline



Advantages of a Linux system

- Risk reduction
- Business needs
- Stability and security
- Flexibility
- Customization
- Support
- Cost reduction

Advantages - Risk reduction

- Changes in the market or customer needs may cause companies to change software frequently
- Can be costly and time-consuming
- Support for closed source software may end
- Vendor may go out of business
- Software version may be retired
- OSS products offer the opportunity to maintain and change the source code

Advantages - Business needs

- Common software available for Linux includes:
- Scientific and engineering software
- Software emulators
- Web servers, web browsers, and e-commerce suites
- Desktop productivity software
- Graphics manipulation software
- Database software
- Security software

Advantages - Stability and security

- Customers using a closed source OS must rely on the OS vendor to fix any bugs
  - Waiting for a hot fix may take weeks or months
- The collaborative open source approach to testing and fixing bugs increases the stability of Linux
- Bugs and security loopholes in OSS programs can be identified and fixed quickly
  - Code is freely available and scrutinized by many developers

Advantages - Flexibility

- Partial list of hardware platforms on which Linux can run:
  - Linux can be (and is) customized to work on mobile and embedded devices

Intel x86/x64	M68K
Itanium	PA-RISC
Mainframe (S/390)	SPARC
ARM	Ultra-SPARC
Alpha	PowerPC
MIPS	

Advantage - Customization

Ability to control the inner workings of the OS

- To use Linux as an Internet Web server, recompile the kernel to include only the support needed to be an Internet Web server
  - Results in a much smaller and faster kernel
- Can choose to install only software packages needed to perform required tasks
- Linux supports several programming languages, such as shell and PERL scripts to customize or automate tasks

Advantage - Support

Linux documentation can be easily found on the Internet

- Frequently asked questions (FAQs)
- HOWTO documents

HOWTO documents are maintained by their authors but are centrally collected by the Linux Documentation Project (LDP)

Linux newsgroups

Linux User Groups (LUGs): Open forum of Linux users who discuss and assist each other in using and modifying the Linux OS

Advantage - Cost reduction

Linux is less expensive than other OSs

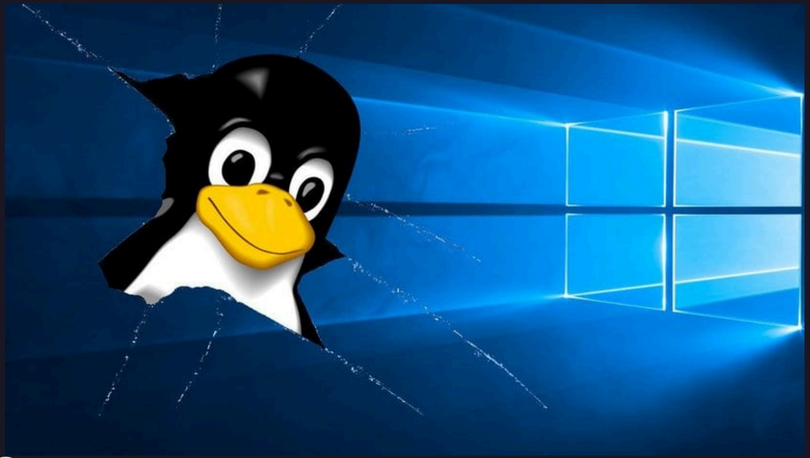
- There is no cost associated with acquiring the software
- A wealth of OSS can run on a variety of different hardware platforms running Linux

The largest costs associated with Linux:

- Costs associated with hiring people to maintain the Linux system
- Potential support from vendors (i.e., Red Hat)

Total cost of ownership (TCO): overall cost of using a particular OS

Disadvantages of a Linux system



No standard edition of Linux

Support means that you'll be searching for very specific fixes

- Browsing SuperUser/ServerFault, forums, etc.

Not as easy to use as closed-source (Windows)

Program support

Common uses of Linux

- Internet servers
- File/print server
- Application server
- Cloud servers
- Supercomputing
- Workstations (Office, personal, scientific, etc.)
- Mobile

Windows

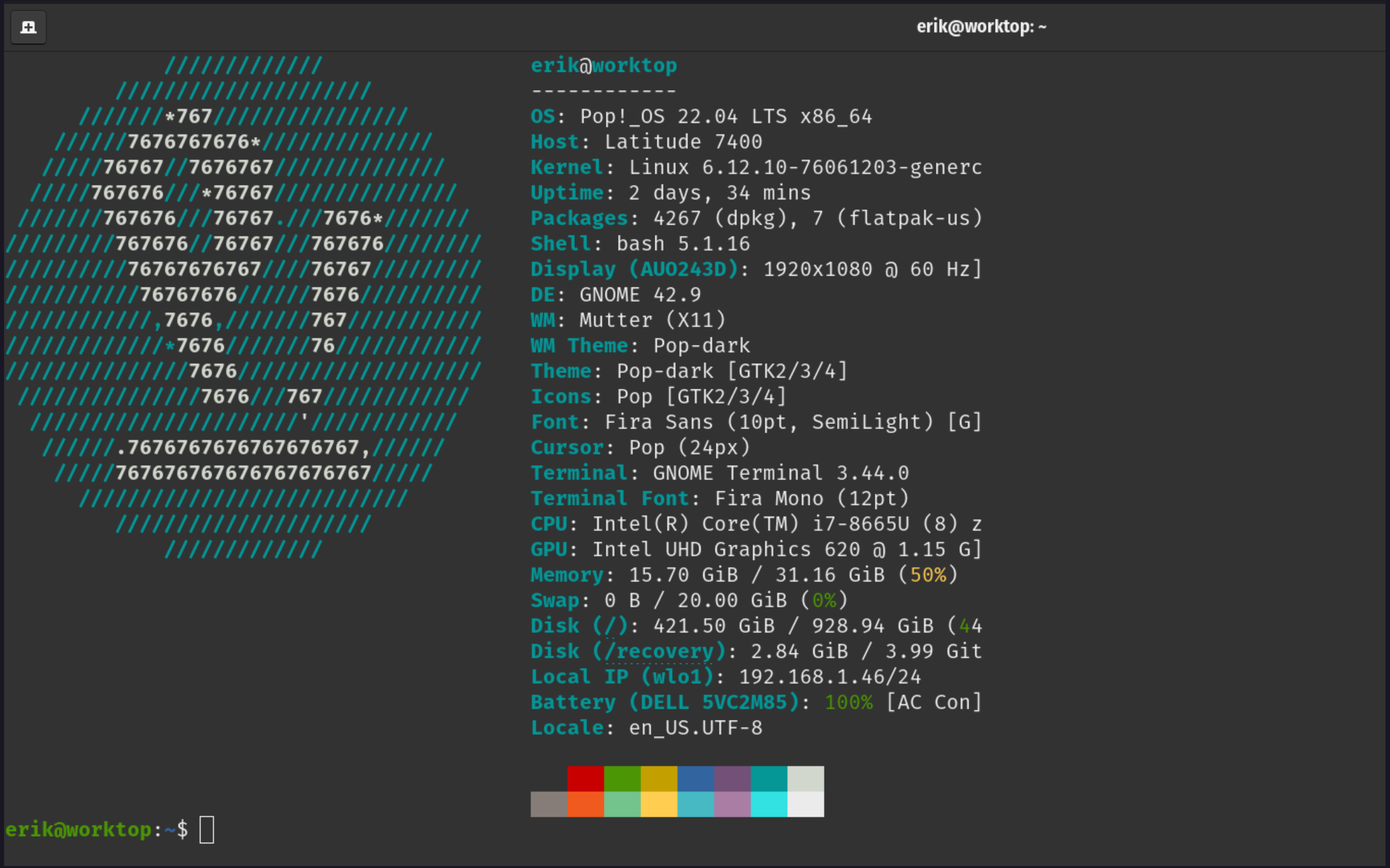
Windows (today), is not Unix-like

- Can you do your homework on a Windows machine?
- Not by default

However, we have WSL! (◕‿◕✿)

- Windows Subsystem for Linux
- If you are on Windows, you must install WSL https://learn.microsoft.com/en-us/windows/wsl/install
    - Be very happy about this - I dreamed about just having the ability to SSH from the terminal when in industry

Terminal


erik@worktop: ~
erik@worktop
OS: Pop!_OS 22.04 LTS x86_64
Host: Latitude 7400
Kernel: Linux 6.12.10-76061203-generc
Uptime: 2 days, 34 mins
Packages: 4267 (dpkg), 7 (flatpak-us)
Shell: bash 5.1.16
Display (AUO243D): 1920x1080 @ 60 Hz]
DE: GNOME 42.9
WM: Mutter (X11)
WM Theme: Pop-dark
Theme: Pop-dark [GTK2/3/4]
Icons: Pop [GTK2/3/4]
Font: Fira Sans (10pt, SemiLight) [G]
Cursor: Pop (24px)
Terminal: GNOME Terminal 3.44.0
Terminal Font: Fira Mono (12pt)
CPU: Intel(R) Core(TM) i7-8665U (8) z
GPU: Intel UHD Graphics 620 @ 1.15 G]
Memory: 15.70 GiB / 31.16 GiB (50%)
Swap: 0 B / 20.00 GiB (0%)
Disk (/): 421.50 GiB / 928.94 GiB (44
Disk (/recovery): 2.84 GiB / 3.99 Git
Local IP (wlo1): 192.168.1.46/24
Battery (DELL 5VC2M85): 100% [AC Con]
Locale: en_US.UTF-8
erik@worktop:~$

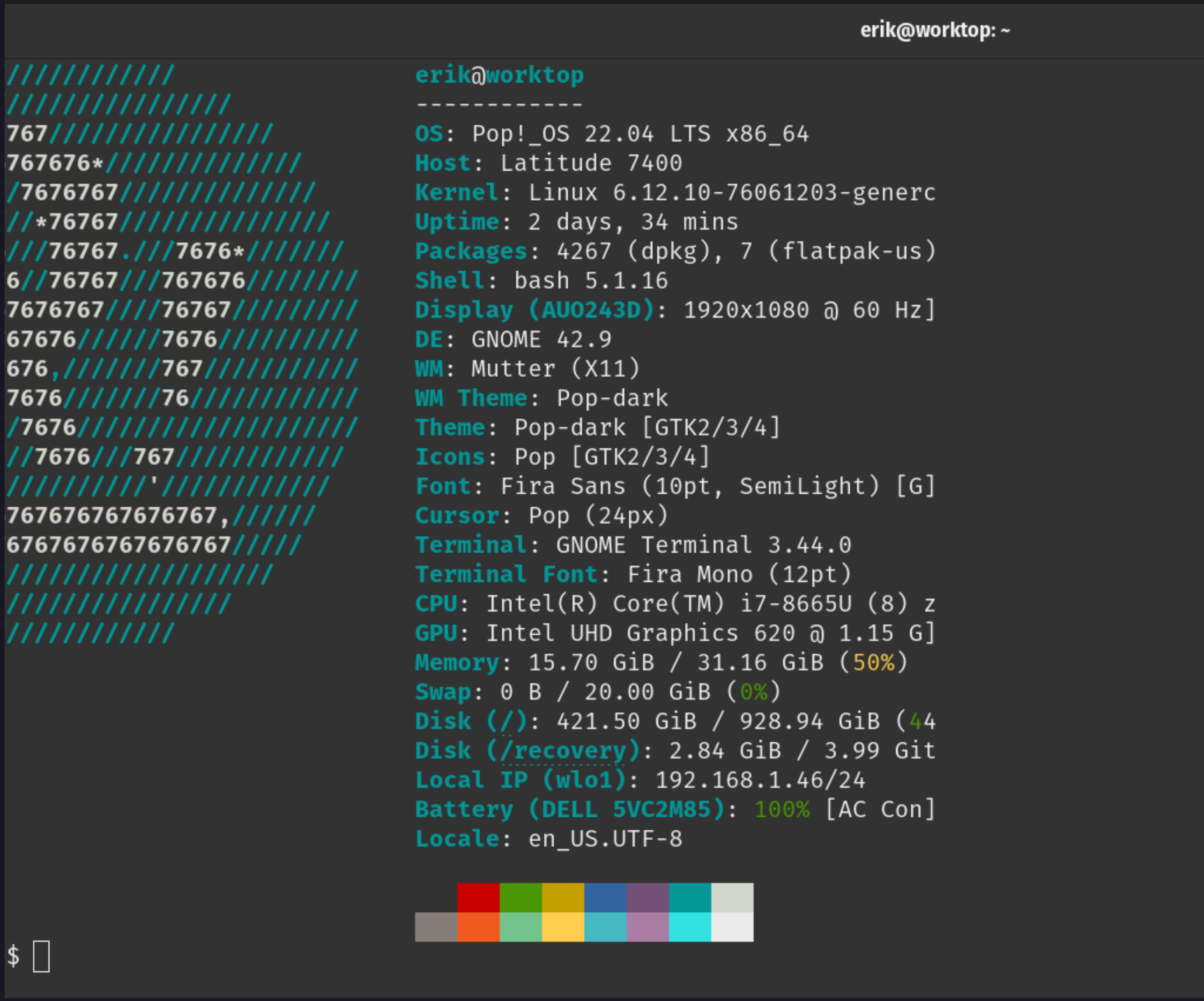
Shells (local)

See the little dollar sign?

Shell!

What shell are we using?

`echo $0`

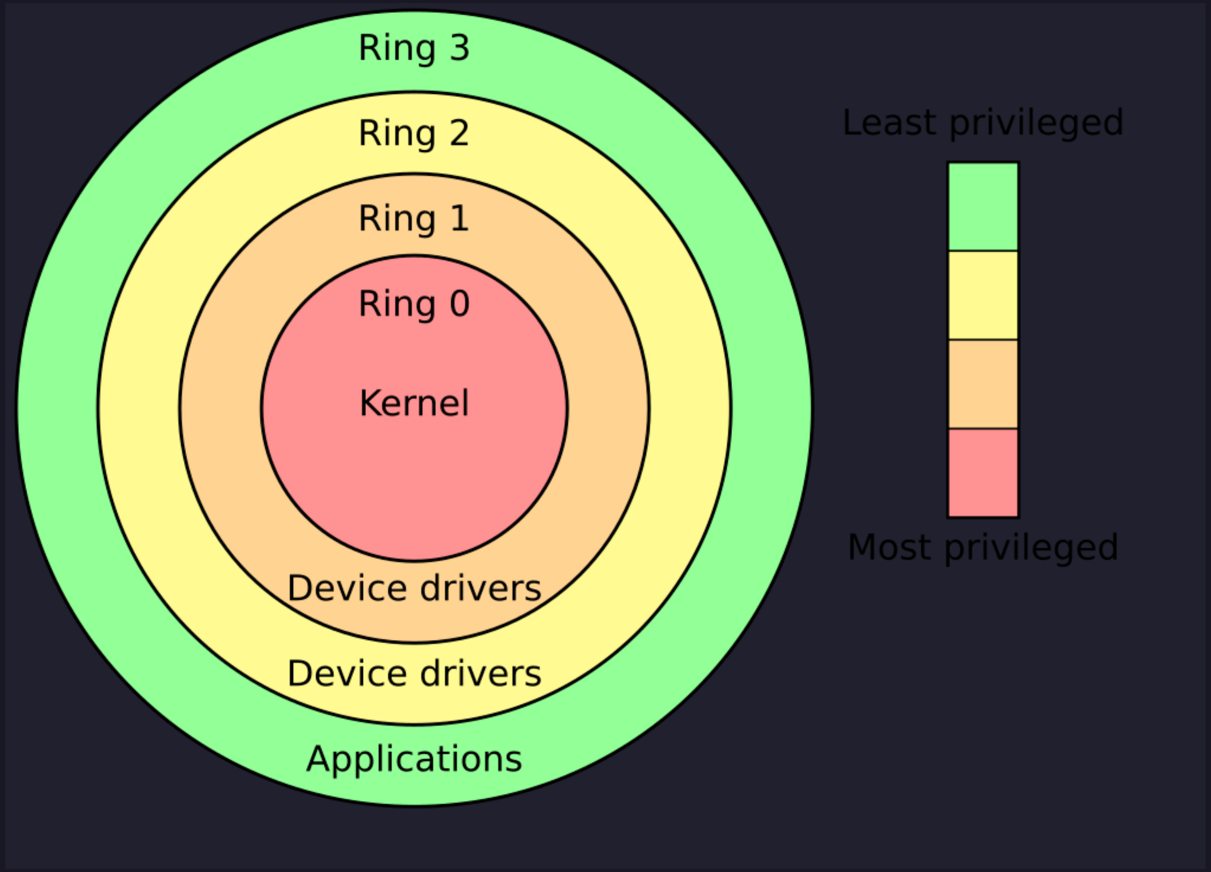


Shells (local)

Where is the shell here?

- Ring 3, typically!

How do we run in other rings?



You can use `eos01.cis.gvsu.edu` - `eos30.cis.gvsu.edu`!

Shells (remote)

Note - if you're off-campus you'll need a VPN!

Or, connecting to another machine

- Or or, **doing your homework** ಠ_ಠ

`ssh` **--> Secure Shell**

Usage: `ssh username@address`

For example, connecting to EOS:

```
ssh username@eos1.cis.gvsu.edu
```

Lab fun

DO NOT

SHUT DOWN

THESE MACHINES

Last - the web interface

https://computing.gvsu.edu

This will provide a web-based point of access for you!

- Allows for desktop or terminal
- Note that the user experience is not great!
  - Can't copy/paste files from your computer to the desktop
  - Random connection breaks common
- However - don't need a VPN!

Ok now really the last

To exit an SSH session (or to kill your terminal)

`exit`